For rest of today: $E \subset \mathbb{R}^n$ is open, $f: E \to \mathbb{R}^n$ is loc. Lip., $I_x \subset \mathbb{R}$ is the maximal interval for any $x \in E$ (w.r.t. $\dot{y} = f(y)$, $y(0) = x$),

$$\Omega = \bigcup_{x \in E} I_x \times \{x\} = \{(t,x) \mid x \in E,\ t \in I_x\} \subset \mathbb{R} \times E,$$

and $\phi: \Omega \to E$ is the flow of f.

Last time $\forall x \in E$: $\dfrac{d}{dt} \underbrace{D\phi^t(x)}_{\frac{\partial}{\partial x}(\phi^t(x))} = Df(\phi^t(x))\, D\phi^t(x)$, $D\phi^0(x) = I$.

Suppose $\bar{x} \in E$ is an equilibrium ($f(\bar{x}) = 0$),

What can we say about $D\phi^t(\bar{x})$?

From above, $\frac{d}{dt} D\phi^t(\bar{x}) = Df(\phi^t(\bar{x}))\, D\phi^t(\bar{x}), \quad D\phi^0(\bar{x}) = I.$

But $\phi^t(\bar{x}) = \bar{x} \quad \forall t \in \mathbb{R}.$

$$\Rightarrow \frac{d}{dt} D\phi^t(\bar{x}) = \underbrace{Df(\bar{x})}_{A}\, D\phi^t(\bar{x}), \quad D\phi^0(\bar{x}) = I$$

Applying our LTI IVP solution formula (& uniqueness thm)

Columnwise $\Rightarrow$

$$\boxed{D\phi^t(\bar{x}) = e^{At} \quad \text{if } \bar{x} \text{ is an equilibrium } [f(\bar{x}) = 0]}.$$

Notation Given $x \in E$, define

forward maximal interval $I_x^+ := I_x \cap [0,\infty)$, backward maximal interval $I_x^- := I_x \cap (-\infty, 0]$

forward orbit $\Gamma_x^+ := \phi^{I_x^+}(x) \subset E$, backward orbit $\Gamma_x^- := \phi^{I_x^-}(x) \subset E$

$= \{\phi^t(x) \mid t \in I_x^+\}$ $= \{\phi^t(x) \mid t \in I_x^-\}$,

orbit $\Gamma_x := \phi^{I_x}(x) = \{\phi^t(x) \mid t \in I_x\} \subset E$

$= \Gamma_x^+ \cup \Gamma_x^-$.

Definition Let $S \subset E$. We say that S is

- forward invariant if $\Gamma_x^+ \subset S$ for all $x \in S$,
- backward invariant if $\Gamma_x^- \subset S$ for all $x \in S$, and
- invariant if S is both forward & backward invariant

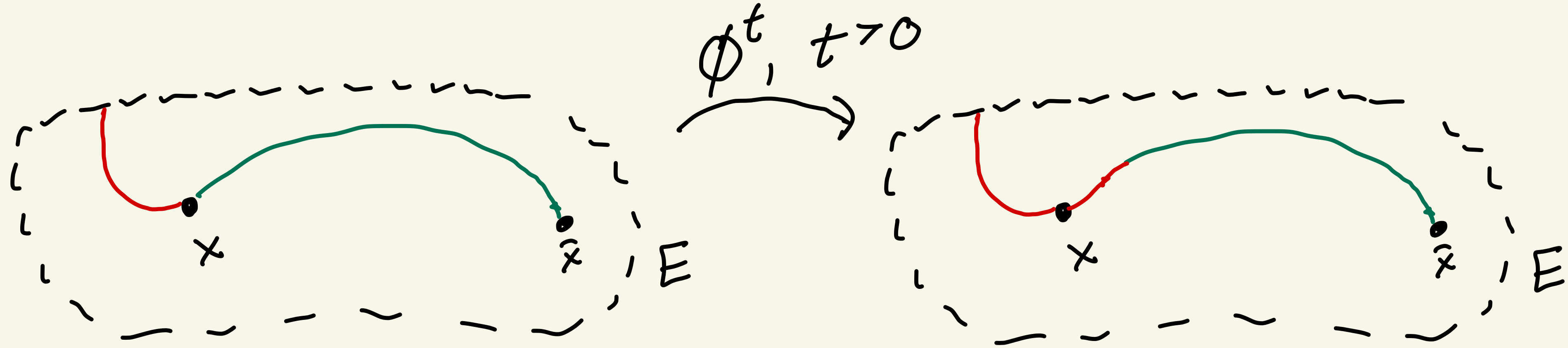
(i.e., $\Gamma_x \subset S$ for all $x \in S$).

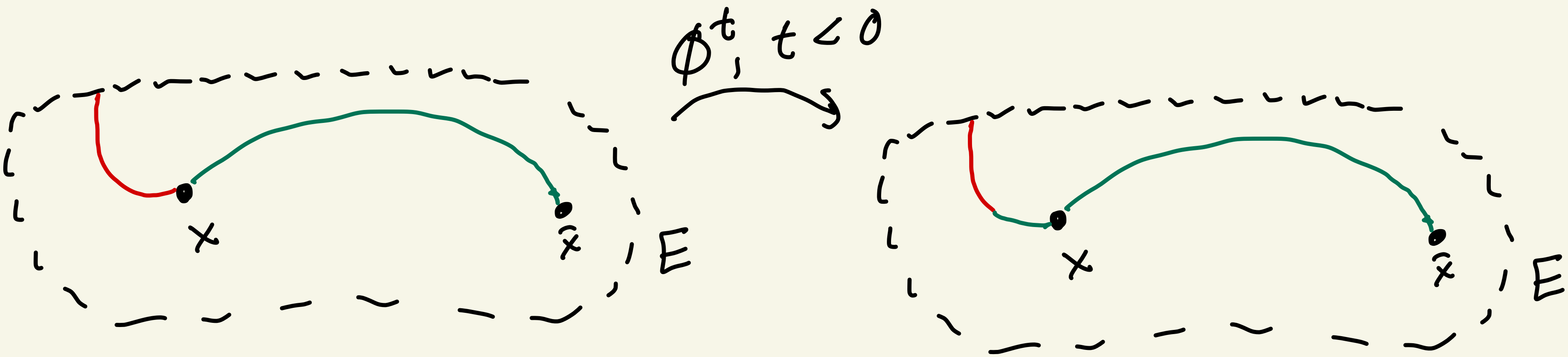
Rmk In other words, S is

- invariant if $\phi^t(x) \in S$ $\forall x \in S$, $t \in I_x$;
- forward invariant if $\phi^t(x) \in S$ $\forall x \in S$, $t \in I_x^+ = I_x \cap [0, \infty)$;
- backward invariant if $\phi^t(x) \in S$ $\forall x \in S$, $t \in I_x^- = I_x \cap (-\infty, 0]$.

Ex Let $\bar{x} \in E$ be an equilibrium. Then $\{\bar{x}\}$ is invariant.

Ex Fix any $x \in E$. Then Γ_x, Γ_x^+, Γ_x^- are respectively invariant, forward invariant, and backward invariant.





Rmk $E = \bigcup_{x \in E} \Gamma_x$, and $\{\Gamma_x\}_{x \in E}$ is in fact a <u>partition</u> of E, since if $\Gamma_x \cap \Gamma_y$ is nonempty then $\Gamma_x = \Gamma_y$.

We have seen that a loc. Lip. f "generates" a loc. Lip flow $\phi: \Omega \to E$ w/ $\Omega \subset \mathbb{R}\times E$ open.

Moreover, w/ $\Omega_t := \{x \in E \mid (t,x) \in \Omega\}$, $\phi^t: \Omega_t \to \Omega_{-t}$ is a homeomorphism. We also saw that $f \in C^k \Rightarrow \phi \in C^k$, and the same argument $\Rightarrow$ $\phi^t: \Omega_t \to \Omega_{-t}$ is a C^k diffeomorphism (C^k bijection w/ C^k inverse).

There is a converse to the fact above; let us just discuss a partial converse to avoid technicalities.

Partial converse Let $\psi: \mathbb{R}\times E \to E$ s.t. (w/ $\psi^t := \psi(t,\cdot)$)

- $\psi^0 = id_E$ (i.e. $\psi^0(x) = x$ $\forall x \in E$).
- $\psi^t \circ \psi^s = \psi^{t+s}$ ($\forall x \in E$, $t,s \in \mathbb{R}$: $\psi^t(\psi^s(x)) = \psi^{t+s}(x)$).
- For any $x \in E$, $t \mapsto \psi^t(x)$ is differentiable.

Then $\exists$ a vector field $g: E \to \mathbb{R}^n$ s.t.

$$\forall x \in E,\ t \in \mathbb{R}: \quad \frac{d}{dt}\psi^t(x) = g(\psi^t(x)).$$

pf Define $g: E \to \mathbb{R}^n$ by

$$g(x) := \frac{d}{ds}\psi^s(x)\Big|_{s=0}.$$

Then

$$\frac{d}{dt}\psi^t(x) = \frac{d}{ds}\psi^{s+t}(x)\Big|_{s=0}$$

$$= \frac{d}{ds}\left(\psi^s(\psi^t(x))\right)\Big|_{s=0}$$

$$= g(\psi^t(x)).$$

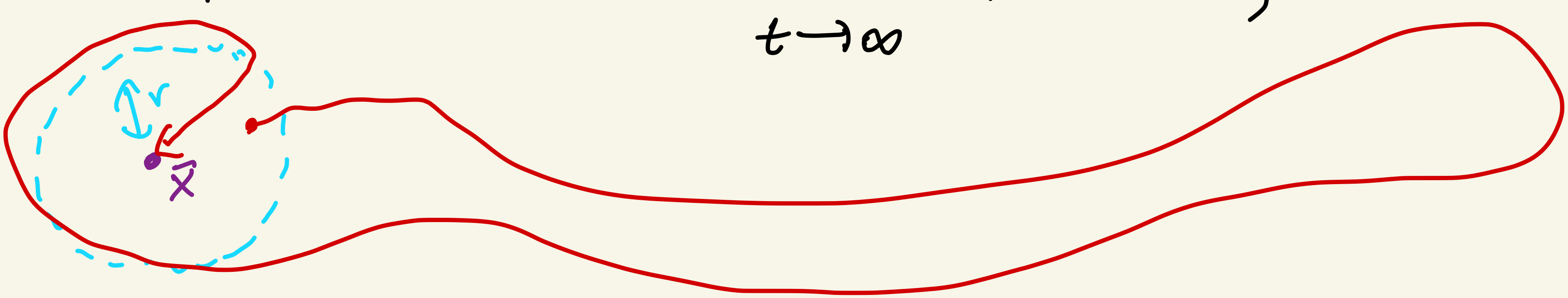
$\square$

Stability definitions

Def Let $\bar{x} \in E$ be an equilibrium. Then $\bar{x}$ is

- **(Lyapunov) stable** if $\forall \varepsilon > 0 \; \exists \delta > 0$ s.t. $\forall x \in \bar{B}_\delta(\bar{x}) \cap E$:
  $\phi^t(x) \in B_\varepsilon(\bar{x})$ for all $t \in I_x^+$;

- **attracting** if $\exists r > 0$ s.t. $\forall x \in B_r(\bar{x}) \cap E$:
  $I_x^+ = [0, \infty)$ and $\lim_{t \to \infty} \phi^t(x) = \bar{x}$;



- **asymptotically stable** if $\bar{x}$ is both (Lyapunov) stable and attracting.